

# **Laboratory Work 1 International Airport Database, ER Diagram**

Course: Databases

Topic: International Airport Database

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## Introduction

The purpose of this laboratory work is to design an Entity-Relationship Diagram for an International Airport database. The system is designed to manage information about airports, airlines, flights, passengers, bookings, boarding passes, baggage, baggage checks and security checks.

### REQUIRED TASKS:

#### TASK 1: Identified entities and brief description

1. AIRPORT: Stores each airport and its location.
2. AIRLINE: Stores airline companies operating flights.
3. FLIGHT: Stores schedule, gates, actual times, airline, and both endpoint airports.
4. PASSENGER: Stores personal and passport information.
5. BOOKING: Connects one passenger to one flight and stores details.
6. BOOKING\_FLIGHT\_CHANGE: Keeps the history of changes to a booking.
7. BOARDING\_PASS: Stores the boarding info generated for a booking.
8. BAGGAGE: Stores individual registered baggage items
9. BAGGAGE\_CHECK: Stores baggage check records
10. SECURITY\_CHECK: Stores security check records

#### TASK 2: Determine attributes

## AIRPORT

| KEY | ATTRIBUTE    | DATA TYPE    | MEANING                   |
|-----|--------------|--------------|---------------------------|
| PK  | airport_id   | INTEGER      | unique airport identifier |
|     | airport_name | VARCHAR(100) | name                      |
|     | country      | VARCHAR(100) | country                   |
|     | state        | VARCHAR(100) | region                    |
|     | city         | VARCHAR(100) | city                      |
|     | created_at   | TIMESTAMP    | creation time             |
|     | updated_at   | TIMESTAMP    | last changed time         |

## AIRLINE

| KEY | ATTRIBUTE    | DATA TYPE    | MEANING                   |
|-----|--------------|--------------|---------------------------|
| PK  | airline_id   | INTEGER      | unique airline identifier |
| UQ  | airline_code | INTEGER      | code                      |
|     | name         | VARCHAR(100) | company name              |
|     | country      | VARCHAR(100) | country                   |
|     | created_at   | TIMESTAMP    | creation time             |

|  |            |             |                         |
|--|------------|-------------|-------------------------|
|  | updated_at | TIMESTAMPTZ | last<br>changed<br>time |
|--|------------|-------------|-------------------------|

## FLIGHT

| KEY | ATTRIBUTE                | DATA TYPE         | MEANING                       |
|-----|--------------------------|-------------------|-------------------------------|
| PK  | flight_id                | INTEGER           | Unique<br>scheduled flight id |
| FK  | airline_id               | INTEGER           | reference<br>s                |
| FK  | departure_airport_id     | INTEGER           | reference<br>s                |
| FK  | arrival_airport_id       | INTEGER           | reference<br>s                |
|     | departing_gate           | VARCHAR(100)<br>) | Departure<br>gate             |
|     | arriving_gate            | VARCHAR(100)<br>) | arrival<br>gate               |
|     | scheduled_departure_time | TIMESTAMP         | planned<br>departure<br>time  |
|     | scheduled_arrival_time   | TIMESTAMP         | planned<br>arrival<br>time    |
|     | actual_departure         | TIMESTAMP         | actual<br>departure<br>time   |

|  |                |           |                     |
|--|----------------|-----------|---------------------|
|  | actual_arrival | TIMESTAMP | actual arrival time |
|  | created_at     | TIMESTAMP | creation time       |
|  | updated_at     | TIMESTAMP | last changed time   |

## PASSENGER

| KEY       | ATTRIBUTE             | DATA TYPE    | MEANING                     |
|-----------|-----------------------|--------------|-----------------------------|
| <b>PK</b> | passenger_id          | INTEGER      | Unique passenger identifier |
|           | first_name            | VARCHAR(100) | first name                  |
|           | last_name             | VARCHAR(100) | last name                   |
|           | gender                | VARCHAR(100) | gender                      |
|           | date_of_birth         | DATE         | birth date                  |
|           | countr_of_citizenship | VARCHAR(100) | citizenship                 |
|           | country_of_residency  | VARCHAR(100) | country lives               |
| <b>UQ</b> | passport_number       | VARCHAR(100) | passport number             |
|           | created_at            | TIMESTAMP    | creation time               |
|           | updated_at            | TIMESTAMP    | last changed time           |

## BOOKING

| KEY | ATTRIBUTE        | DATA TYPE     | MEANING                   |
|-----|------------------|---------------|---------------------------|
| PK  | booking_id       | INTEGER       | Unique booking identifier |
| FK  | flight_id        | INTEGER       | References                |
| FK  | passenger_id     | INTEGER       | References                |
|     | status           | VARCHAR(100)  | Booking state             |
|     | booking_platform | VARCHAR(100)  | platform made booking     |
|     | ticket_price     | NUMERIC(12,2) | booking price             |
|     | created_at       | TIMESTAMP     | creation time             |
|     | updated_at       | TIMESTAMP     | last changed time         |

## BOOKING\_CHANGE

| KEY | ATTRIBUTE     | DATA TYPE | MEANING                  |
|-----|---------------|-----------|--------------------------|
| PK  | change_id     | INTEGER   | Unique change identifier |
| FK  | booking_id    | INTEGER   | References               |
| FK  | old_flight_id | INTEGER   | References               |
| FK  | new_flight_id | INTEGER   | References               |
|     | changed_time  | TIMESTAMP | Time the change occurred |

## BOARDING\_PASS

| KEY | ATTRIBUTE        | DATA TYPE    | MEANING                         |
|-----|------------------|--------------|---------------------------------|
| PK  | boarding_pass_id | INTEGER      | Unique boarding pass identifier |
| FK  | booking_id       | INTEGER      | References                      |
|     | seat             | VARCHAR(100) | Assigned seat                   |
|     | boarding_time    | TIMESTAMP    | time                            |
|     | created_at       | TIMESTAMP    | creation time                   |
|     | updated_at       | TIMESTAMP    | last changed time               |

## BAGGAGE

| KEY | ATTRIBUTE    | DATA TYPE    | MEANING                        |
|-----|--------------|--------------|--------------------------------|
| PK  | baggage_id   | INTEGER      | Unique baggage item identifier |
| FK  | booking_id   | INTEGER      | References                     |
|     | weight_in_kg | NUMERIC(6,2) | Item weight in kilograms       |
|     | created_at   | TIMESTAMP    | creation time                  |
|     | updated_at   | TIMESTAMP    | last changed time              |

## SECURITY\_CHECK

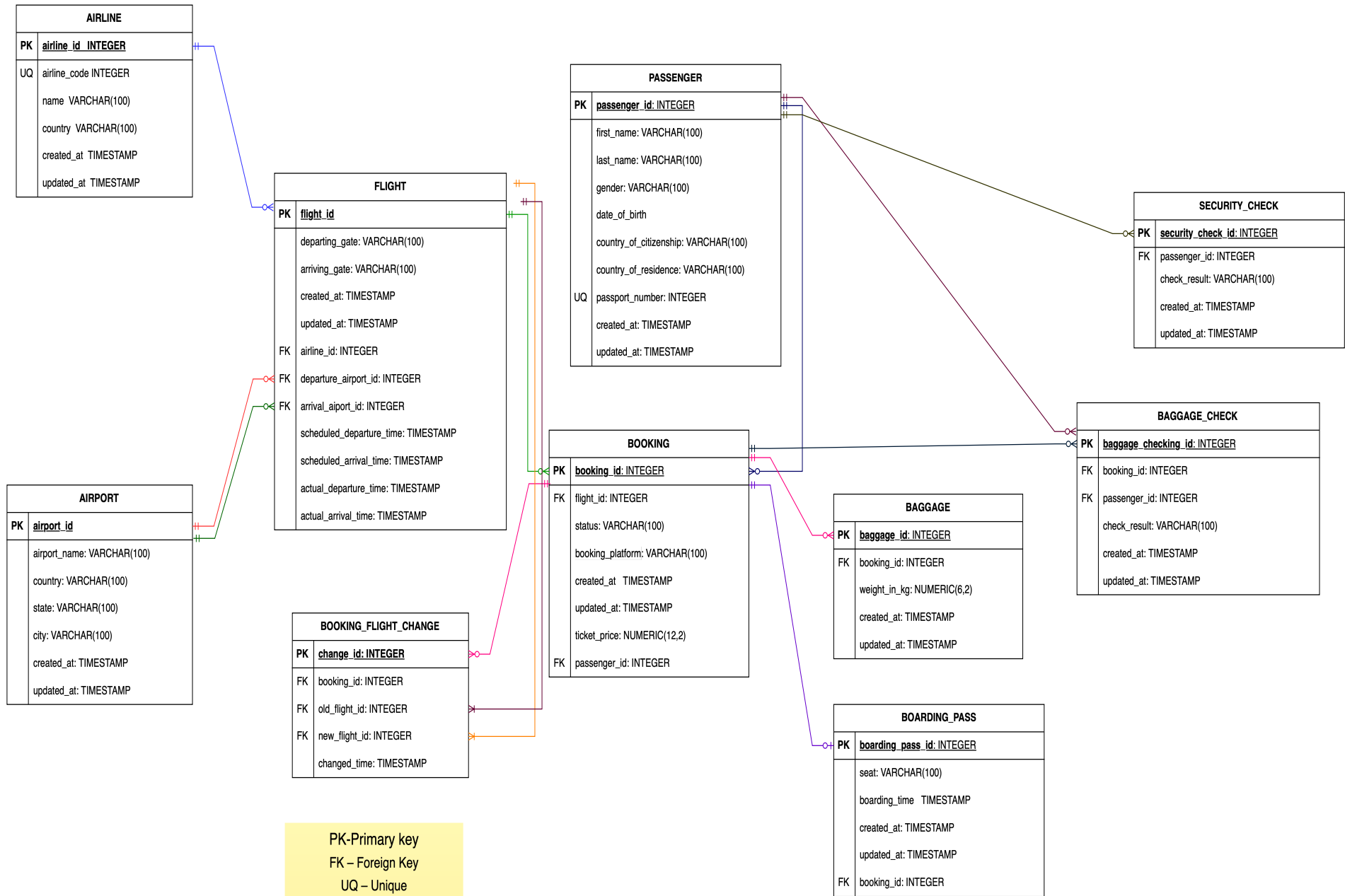
| KEY | ATTRIBUTE         | DATA TYPE    | MEANING                          |
|-----|-------------------|--------------|----------------------------------|
| PK  | security_check_id | INTEGER      | Unique security check identifier |
| FK  | passenger_id      | INTEGER      | References                       |
|     | check_result      | VARCHAR(100) | check outcome                    |

|  |            |           |                   |
|--|------------|-----------|-------------------|
|  | created_at | TIMESTAMP | creation time     |
|  | updated_at | TIMESTAMP | last changed time |
|  |            |           |                   |
|  |            |           |                   |

## BAGGAGE\_CHECK

| KEY | ATTRIBUTE           | DATA TYPE    | MEANING                         |
|-----|---------------------|--------------|---------------------------------|
| PK  | baggage_checking_id | INTEGER      | Unique baggage check identifier |
| FK  | booking_id          | INTEGER      | References                      |
| FK  | passenger_id        | INTEGER      | References                      |
|     | check_result        | VARCHAR(100) | check outcome                   |
|     | created_at          | TIMESTAMP    | creation time                   |
|     | updated_at          | TIMESTAMP    | last changed time               |

## Entity Relationship Diagram



TASK : Identify relationships

## Cardinality and Participation Constraints

1 = exactly one ,mandatory

0..1 = zero or one ,optional

1..N = one or many,mandatory

0..N = zero or many,optional

AIRLINE-**operates**-FLIGHT: 1:0..N **Airline** may have no/many flights; **Flight** must have one **Airline**

AIRPORT-**of departure**-FLIGHT(DEPARTURE): 1:0..N **Airport** may be departure point for no/many flights; **Flight** must have one departure airport

AIRPORT-**of arrival**-FLIGHT(ARRIVAL): 1:0..N **Airport** may be arrival point for no/many flights; **Flight** must have one arrival airport

FLIGHT-**has**-BOOKING: 1:0..N **Flight** may have no/many bookings; **Booking** must refer to one flight

PASSENGER-**make**-BOOKING: 1:0..N **Passenger** may have no/many bookings; **Booking** must belong to one passenger

BOOKING-**register**-BAGGAGE: 1:0..N **Booking** may have no/many bags; **Baggage** must belong to one booking

BOOKING-produce-BOARDING\_PASS: 1:0..1 **Booking may have no boarding pass yet; Boarding Pass must belong to one booking**

PASSENGER-passes-SECURITY\_CHECK: 1:0..N **Passenger may have multiple checks; each check belongs to one passenger**

BOOKING-has-BAGGAGE\_CHECK:1:0..N **Booking may have multiple baggage checks**

PASSENGER-has-BAGGAGE\_CHECK: 1:0..N **Passenger may be associated with multiple baggage checks**

BOOKING-records old/new-BOOKING\_FLIGHT\_CHANGE: 1:0..N **Booking may have multiple changes;each change belongs to one booking.**

## **Conclusion**

**In this laboratory work, an ERD for an International Airport database was designed. The main entities, their attributes, primary keys and foreign keys were identified.**